

Volleyball Canada Registration Database Project

Final Group Project – DBAS 1007

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DBAS 1007 – Data Fundamentals

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1.Introduction

The purpose of this project is to design a well-structured relational database based on the Volleyball Canada registration form. The original form collects a wide range of information about athletes, parents or guardians, training preferences, and previous volleyball experience. However, because all of this information is stored together on a single form, it is not suitable for efficient data storage or long-term management. To address this problem, our group transformed the form into a normalized database that can store the information more clearly, reduce redundancy, and support future data analysis.

In this project, we identify the main entities from the form, define their relationships, and develop an Entity Relationship Diagram (ERD) using Crow's Foot notation. In addition, we create a complete data dictionary that explains each table and attribute, including data types and constraints. Finally, we produce the SQL scripts needed to build the database. Through this process, we aim to demonstrate our understanding of normalization principles and to create a functional database structure that could be used by a real volleyball organization to manage athlete registrations more effectively.

2. Project Form (Original Registration Form)



Volleyball Canada Regional Excellence Program VC REP - Okanagan (2022-2023)



ATHLETE'S INFORMATION

First Name: Last Name:

Date of Birth: (dd/mm/year) Gender: Grade:

Phone: Cell: Email: Main communication email

Address: Postal Code:

Parent's Name: Cell: Email:

I am registering for the following program (please check one):

a) 2 Days per week: ☐ b) 1 Day per week: ☐ T-shirt size: s/m/l/xl

Preferred Training Day 1st choice for Training Day 2nd choice for Training Day 3rd choice for Training Day (if applicable)

VOLLEYBALL EXPERIENCE

School: Position: Coach:

Club: Position: Coach:

Other teams that you have played on, Team BC, BC Summer Games, etc:

PARENT/GUARDIAN RELEASE

I hereby authorize my child's participation in the Volleyball Canada Regional Excellence Program (VC REP). I know of no physical or mental problems which may affect my child's ability to safely participate in this program. I hereby authorize the staff of the VC REP to act on my behalf in the case of illness or injury involving my child. I agree that VC REP and/or its instructor(s), agents, employees, servants or any of them, shall not be held liable for any injuries or damages which may arise out of the aforementioned activities, regardless of cause, unless such injuries or damages result expressly from the sole negligence of the VC REP staff, its instructor(s), agents, employees and servants while acting within the scope of their duties. By agreeing to this release, it is my intention to exempt and relieve the VC REP, its instructors and employees, agents and servants from any and all liability for personal injury, property damage and wrongful death. I am aware that the VC REP staff does not provide medical/accident insurance for the enrolled participant and I understand that the responsibility to arrange such insurance, or to otherwise cover any medical costs, is mine. VC REP may occasionally take pictures of its participants for use in promotional, advertisement materials or publications (brochures, websites, social media, newspaper ads, etc.). By signing below, you agree to allow VC REP to reproduce the likeness of your child in such promotional/ advertisement materials and publications. I, as the Parent/Guardian, agree to the terms and conditions above:

Name: Signature: Date:

Please submit your registration via email to: vcprep.ok@gmail.com



Volleyball Canada Regional Excellence Program Okanagan



For more information contact: vcrep.ok@gmail.com or Steve Manuel: steve.manuel@ubc.ca

The Volleyball Canada Regional Excellence Program is a high performance volleyball training program run by a professional coaching staff, including UBC Okanagan Varsity coaches and athletes. The Volleyball Canada Regional Excellence Program Head Coaches will lead the training sessions including core essential development curriculum, position specific training, small group training, assessment/evaluation, and preparing/identifying athletes for the next level of performance.

What does the Okanagan VC REP Program look like?

PROGRAM FEATURES INCLUDE:

- Low Coach to Athlete Ratio - Individualized attention for each athlete with an emphasis on skill development.
- Position Specific Development - Unique & customized for each athlete.
- Video Analysis - Evaluation and analysis of each athlete's technique and positioning.
- Strength & Conditioning - Introduction and emphasis on the importance of off court training.
- Flexible Training Model - Consideration for academic schedules (semester or linear scholastic models).
- Complementary Training - Minimal interference with high school and club seasons (training sessions traditionally do not interfere with high school or club programs).
- Team BC - Ignite, BC Summer Games, Western Canada Summer Games, and Canada Summer Games Pathway.
- Team Canada - Youth, Junior, and National Team Pathway.

Annual Program Fees:

2 Days per week program - \$1500 (includes all above features and 2 training sessions per week for 22 weeks)

1 Day per week program - \$950 (includes all above features and 1 training session per week for 22 weeks)

September 12, 2022 – December 1, 2022 (Monday to Thursday; 7:00am – 9:00am)

AND

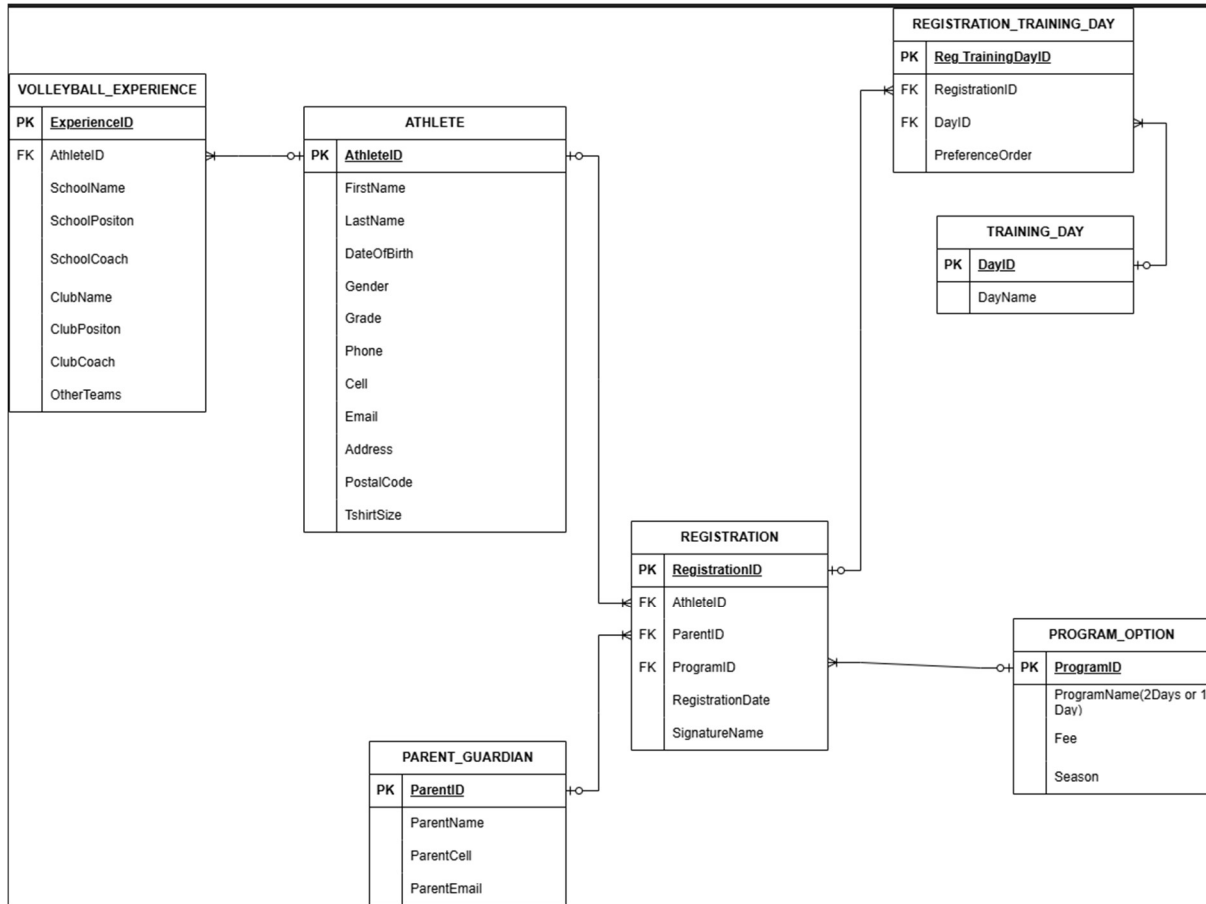
January 9, 2023 – March 30, 2023 (Monday to Thursday; 7:00am – 9:00am)

*****NOTE: There will be no VC REP training on October 10, 2022, November 7-10, 2022, February 20, 2023, or March 20-23, 2023.**

Space is limited, please register early.

Please submit your registration via email to: vcrep.ok@gmail.com

3.Entity Relationship Diagram (ERD)



4.Data Dictionary

The screenshot below shows the full Excel Data Dictionary. The original Excel file has also been attached separately.

[illegible]

We separated the full Excel Data Dictionary into individual tables and created a table for each one.

4.1. Table: ATHLETE

TABL E NAM E	ATTRI BUTE NAME	CONT ENTS	TYPE	FORMA T	RANGE	REQU IRED	P K O R F K	FK REFERE NCED TABLE
ATHL ETE	Athlete ID	Unique athlete identifi er	INT	9999	1–9999	Y	P K	
ATHL ETE	FirstNa me	Athlete first name	VARCH AR(30)	Xxxxx		Y		
ATHL ETE	LastNa me	Athlete last name	VARCH AR(30)	Xxxxx		Y		
ATHL ETE	Date Of Birth	Date of birth	DATE	yyyy- mm-dd		Y		

ATHL ETE	Gender	Gender	VARCH AR(10)	Xxxx	Male/Female/Other	Y		
ATHL ETE	Grade	School grade	INT	99	1–12	Y		
ATHL ETE	Phone	Phone number	VARCH AR(15)	999-999- 9999		N		
ATHL ETE	Cell	Cell phone	VARCH AR(15)	999-999- 9999		N		
ATHL ETE	Email	Email address	VARCH AR(50)	Xxxx@x xxx.xx		N		
ATHL ETE	Address	Home address	VARCH AR(60)	Xxxxxxxx xx		N		
ATHL ETE	Postal Code	Postal code	VARCH AR(10)	A9A 9A9	Valid Canadian postal code	N		
ATHL ETE	Tshirt Size	T-shirt size	VARCH AR(5)	Xx	XS/S/M/L/ XL	N		

4.2. Table: PARENT_GUARDIAN

TABLE NAME	ATTRI BUTE NAME	CONTE NTS	TYPE	FORMA T	RA NG E	REQU IRED	P K	FK REFERE NCED TABLE
PARENT_G UARDIAN	Parent ID	Unique parent/g uardian identifier	INT	9999	1– 999 9	Y	P K	
PARENT_G UARDIAN	Parent Name	Parent/ guardian name	VARCH AR(40)	Xxxxx Xxxxx		Y		
PARENT_G UARDIAN	Parent Cell	Parent/g uardian cell phone	VARCH AR(15)	999-999- 9999		N		
PARENT_G UARDIAN	ParentE mail	Parent/g uardian email	VARCH AR(50)	Xxxx@x xxx.xx		N		

4.3. Table: PROGRAM_OPTION

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	FOR MAT	RANGE	REQUIRED	PK OR FK	FK REFERENCED TABLE
PROGRAM_ OPTION	Program ID	Unique program identifier	INT	99	1–99	Y	PK	
PROGRAM_ OPTION	Program Name	Program name	VARCHAR(40)	Xxxxx xxxxx		Y		
PROGRAM_ OPTION	Fee	Program fee	DECIMAL(7,2)	9,999. 99	0– 9,99 9.99	Y		
PROGRAM_ OPTION	Season	Season (e.g., 2022– 2023)	VARCHAR(9)	9999– 9999	2000 – 2100	Y		

4.4. Table: REGISTRATION

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	FORMAT	RANGE	REQUIRED	PK	FK REFERENCE D TABLE
REGISTRATION	RegistrationID	Unique registration ID	INT	99999	1–99999	Y	PK	
REGISTRATION	AthleteID	FK to Athlete	INT	9999	1–9999	Y	FK	ATHLETE
REGISTRATION	ParentID	FK to Parent	INT	9999	1–9999	Y	FK	PARENT_GUARDIAN
REGISTRATION	ProgramID	FK to Program Option	INT	99	1–99	Y	FK	PROGRAM_OPTION
REGISTRATION	RegistrationDate	Date submitted	DATE	yyyy-mm-dd		Y		
REGISTRATION	Signature	Signer	VARCH	Xxxxx		Y		

ATION	Name	name	AR(50)	Xxxxx				
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4.5. Table: TRAINING_DAY

TABLE NAME	ATTRI BUTE NAME	CONTE NTS	TYPE	FOR MAT	RAN GE	REQUI RED	P K	FK REFERE NCED TABLE
TRAININ G_DAY	DayID	Trainin g day ID	INT	9	1–7	Y	P K	
TRAININ G_DAY	DayNa me	Trainin g day name	VARCH AR(15)	Xxxx xxxx	Mond ay– Sund ay	Y		

4.6. Table: REGISTRATION_TRAINING_DAY

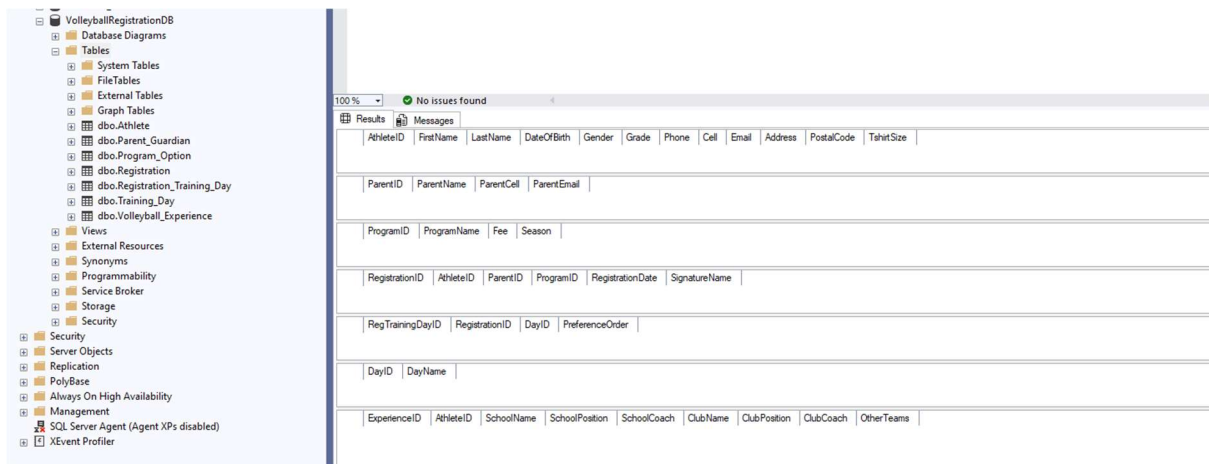
TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	FORMAT	RANGE	REQUIRED	PK	FK REFERENCED TABLE
REGISTRATION_TRAINING_DAY	RegTrainingDayID	Unique row ID	INT	99999	1–99999	Y	PK	
REGISTRATION_TRAINING_DAY	RegistrationID	FK to Registration	INT	99999	1–99999	Y	FK	REGISTRATION
REGISTRATION_TRAINING_DAY	DayID	FK to Training Day	INT	9	1–7	Y	FK	TRAINING_DAY
REGISTRATION_TRAINING_DAY	Preference Order	Preference order	INT	9	1–3	Y		

4.7. Table: VOLLEYBALL_EXPERIENCE

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	FOR MAT	RANGE	REQUIRED	PK OR FK	FK REFERENCED TABLE
VOLLEYBALL_EXPERIENCE	ExperienceID	Unique experience ID	INT	99999	1–99999	Y	PK	
VOLLEYBALL_EXPERIENCE	AthleteID	FK to Athlete	INT	9999	1–9999	Y	FK	ATHLETE
VOLLEYBALL_EXPERIENCE	SchoolName	School team name	VARCHAR(50)	Xxxxxxxx		N		
VOLLEYBALL_EXPERIENCE	SchoolPosition	Position at school	VARCHAR(30)	Xxxxxxxx		N		
VOLLEYBALL_EXPERIENCE	SchoolCoach	School coach	VARCHAR(30)	Xxxxxxxx		N		

		name	AR(50)					
VOLLEYBALL _EXPERIENCE	ClubName	Club team name	VA RCH AR(50)	Xxxx xxxxx		N		
VOLLEYBALL _EXPERIENCE	ClubPositi on	Club position	VA RCH AR(30)	Xxxx xxx		N		
VOLLEYBALL _EXPERIENCE	ClubCoac h	Club coach name	VA RCH AR(50)	Xxxx xxx		N		
VOLLEYBALL _EXPERIENCE	OtherTea ms	Additional teams played	VA RCH AR(100)	Xxxx xxxxx		N		

5.Database Scripts



-- 1. DATABASE CREATION --

```
CREATE DATABASE VolleyballRegistrationDB;
```

```
GO
```

```
USE VolleyballRegistrationDB;
```

```
GO
```

-- 2. TABLE CREATION --

-- ATHLETE TABLE

```
CREATE TABLE Athlete (
```

```
    AthleteID INT PRIMARY KEY,
```

```
    FirstName VARCHAR(30) NOT NULL,
```

```
    LastName VARCHAR(30) NOT NULL,
```

```
    DateOfBirth DATE NOT NULL,
```

```
    Gender VARCHAR(10) NOT NULL,
```

```
    Grade INT NOT NULL,
```

```
    Phone VARCHAR(15),
```

```
    Cell VARCHAR(15),
```

```
Email VARCHAR(50),
Address VARCHAR(60),
PostalCode VARCHAR(10),
TshirtSize VARCHAR(5)
);

-- PARENT / GUARDIAN TABLE
CREATE TABLE Parent_Guardian (
    ParentID INT PRIMARY KEY,
    ParentName VARCHAR(40) NOT NULL,
    ParentCell VARCHAR(15),
    ParentEmail VARCHAR(50)
);

-- PROGRAM OPTION TABLE
CREATE TABLE Program_Option (
    ProgramID INT PRIMARY KEY,
    ProgramName VARCHAR(40) NOT NULL,
    Fee DECIMAL(7,2) NOT NULL,
    Season VARCHAR(9) NOT NULL
);

-- TRAINING DAY TABLE
CREATE TABLE Training_Day (
    DayID INT PRIMARY KEY,
    DayName VARCHAR(15) NOT NULL
);

-- REGISTRATION TABLE
CREATE TABLE Registration (
```

```

RegistrationID INT PRIMARY KEY,
AthleteID INT NOT NULL,
ParentID INT NOT NULL,
ProgramID INT NOT NULL,
RegistrationDate DATE NOT NULL,
SignatureName VARCHAR(50) NOT NULL
);

```

```
-- REGISTRATION TRAINING DAY TABLE
```

```

CREATE TABLE Registration_Training_Day (
    RegTrainingDayID INT PRIMARY KEY,
    RegistrationID INT NOT NULL,
    DayID INT NOT NULL,
    PreferenceOrder INT NOT NULL
);

```

```
-- VOLLEYBALL EXPERIENCE TABLE
```

```

CREATE TABLE Volleyball_Experience (
    ExperienceID INT PRIMARY KEY,
    AthleteID INT NOT NULL,
    SchoolName VARCHAR(50),
    SchoolPosition VARCHAR(30),
    SchoolCoach VARCHAR(50),
    ClubName VARCHAR(50),
    ClubPosition VARCHAR(30),
    ClubCoach VARCHAR(50),
    OtherTeams VARCHAR(100)
);

```

```
-- 3 & 4. FOREIGN KEYS WITH CASCADE RULES --
```

-- Registration → Athlete

ALTER TABLE Registration

ADD CONSTRAINT FK_Registration_Athlete

FOREIGN KEY (AthleteID)

REFERENCES Athlete(AthleteID)

ON UPDATE CASCADE

ON DELETE CASCADE;

-- Registration → Parent_Guardian

ALTER TABLE Registration

ADD CONSTRAINT FK_Registration_Parent

FOREIGN KEY (ParentID)

REFERENCES Parent_Guardian(ParentID)

ON UPDATE CASCADE

ON DELETE CASCADE;

-- Registration → Program_Option

ALTER TABLE Registration

ADD CONSTRAINT FK_Registration_Program

FOREIGN KEY (ProgramID)

REFERENCES Program_Option(ProgramID)

ON UPDATE CASCADE

ON DELETE CASCADE;

-- Registration_Training_Day → Registration

ALTER TABLE Registration_Training_Day

ADD CONSTRAINT FK_RegTrainingDay_Registration

FOREIGN KEY (RegistrationID)

REFERENCES Registration(RegistrationID)

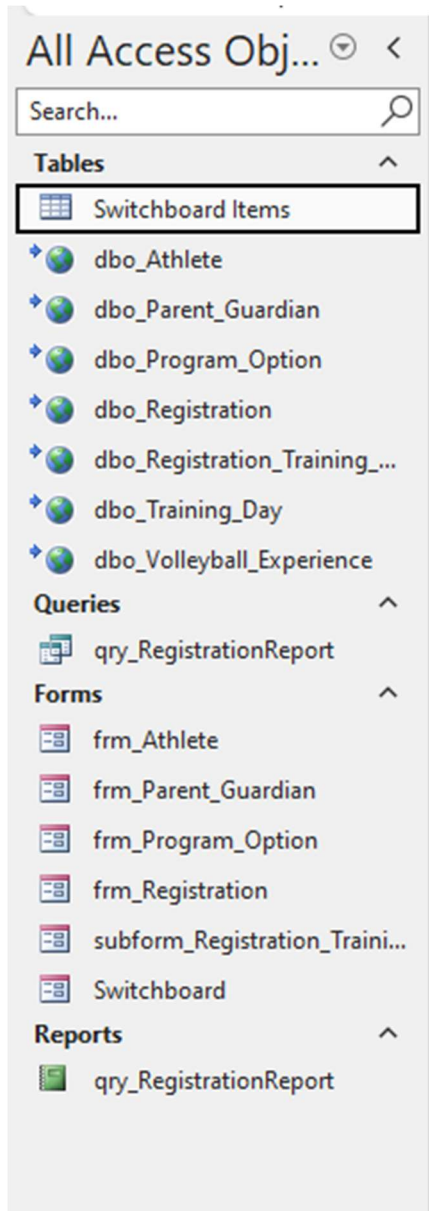
```
ON UPDATE CASCADE  
ON DELETE CASCADE;
```

```
-- Registration_Training_Day → Training_Day  
ALTER TABLE Registration_Training_Day  
ADD CONSTRAINT FK_RegTrainingDay_TrainingDay  
FOREIGN KEY (DayID)  
REFERENCES Training_Day(DayID)  
ON UPDATE CASCADE  
ON DELETE CASCADE;
```

```
-- Volleyball_Experience → Athlete  
ALTER TABLE Volleyball_Experience  
ADD CONSTRAINT FK_VolleyballExperience_Athlete  
FOREIGN KEY (AthleteID)  
REFERENCES Athlete(AthleteID)  
ON UPDATE CASCADE  
ON DELETE CASCADE;
```

```
-- END OF SCRIPT --
```

6. Microsoft Access & SQL Server Interoperability



dbo_Athlete X

Athlete details

AthleteID	
FirstName	
LastName	
DateOfBirth	
Gender	v
Grade	
Phone	
Cell	
Email	
Address	
PostalCode	
TshirtSize	

frm_Parent_Guardian ×

Parent/Guardian details

▶

ParentID	
ParentName	
ParentCell	
ParentEmail	

frm_Program_Option ×

Program Options

▶

ProgramID	
ProgramName	
Fee	
Season	

dbo_Registration X

Program Registration

Registration ID

Athlete ID

Program ID

Date

Signature Name

Training days

Select how many days per week

Day PreferenceOrder

Record: 1 of 1 No Filter Search

subform_Registration_Training_Day X

Select how many days per week

Day PreferenceOrder

Athlete Registration

Athlete Name

ProgramName

Registration ID

Registration Date

AthleteID

Season

Days

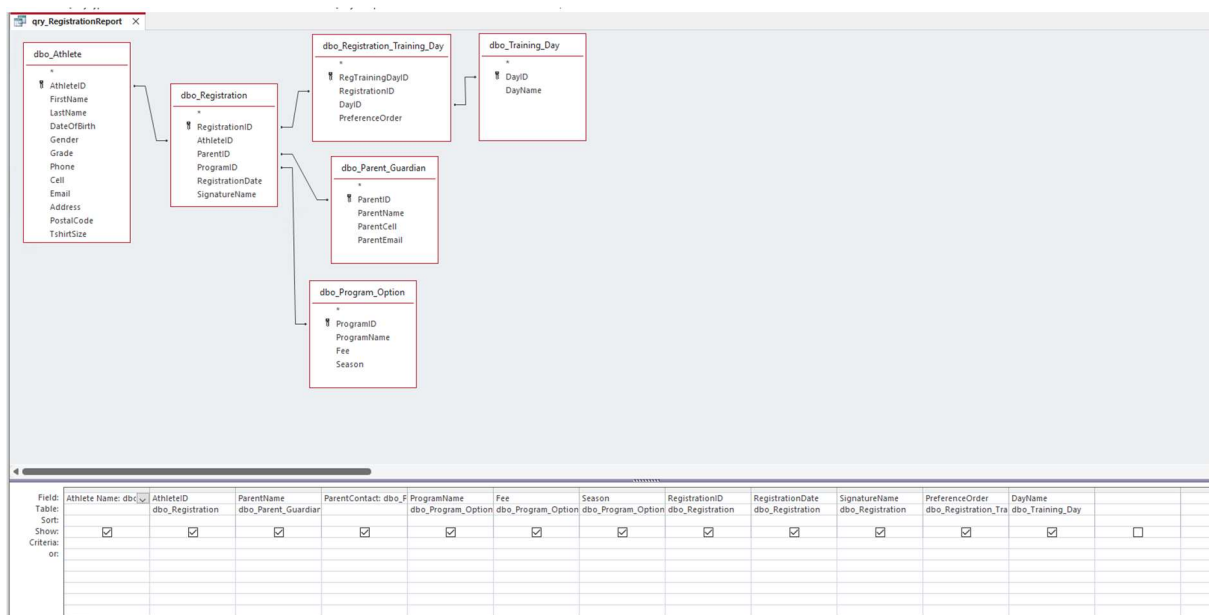
PreferenceOrder

Fee

ParentName

ParentContact

Signature Name





7. Conclusion

This project allowed our group to apply the full database development process from start to finish. We began by analyzing the original registration form and identifying the required entities, attributes, and relationships. Based on this information, we designed an Entity Relationship Diagram (ERD) that clearly represented how athletes, parents, program options, training days, and registration details were connected.

After finalizing the ERD, we created a complete data dictionary that defined each table, field, data type, and key. This step helped us maintain consistency and ensured accuracy when writing the database scripts. We then implemented the database in SQL Server using structured CREATE TABLE statements with proper primary and foreign keys. Insert statements were added to test the system and confirm that relationships and constraints were functioning correctly.

Finally, we demonstrated interoperability between Microsoft Access and SQL Server by linking all SQL tables through an ODBC connection. Forms, queries, and reports were tested to confirm that the front-end could retrieve and display data from the SQL database. Through these steps, we were able to show a working end-to-end system that supports the registration workflow.

Overall, this project strengthened our understanding of relational database design, SQL development, and front-end integration. It also improved our teamwork skills, as each member contributed to different parts of the project. By completing all components, we successfully met the project requirements and gained practical experience that will be valuable for future courses and real-world applications.

8.References

Coronel, C., & Morris, S. (2019). Database systems: Design, implementation, & management (13th ed.). Cengage Learning.

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Project Documentation (20pts)	Entity Relationship Diagram (15pts)			Data Dictionary(5pts)	
Scripts (45pts)	Database Creation (2pts)	Table Creation (20pts)	PK/FK (10pts)	Relationships (10pts)	Cascading Updates and Deletes (3pts)
MS Access- MS SQL Interoperability (20)	ODBC Conn (2pts)	Dashboard (4pts)	Forms (Add/Edit) (8pts)	Report (4pts)	Query (2pts)
Presentation (15pts)	Clearly described ERD and Data Dictionary (5pts)	Execution of DDL/DML Worked (5pts)		MS Access Data Input, Reporting, Query (5pts)	

Notes: